

Huanqi Wen

+8613546387712 | wenhq7@mail2.sysu.edu.cn | huanqiwen.github.io

EDUCATION

- **Sun Yat-sen University** Guangzhou, China
M.S. in Mathematics 09/2023 - present
 - **GPA:** 3.9/4 (89/100)
 - **Advisor:** Prof. Lei Song
 - **Master's Thesis:** The N_p property of line bundles on toric varieties
- **Sun Yat-sen University** Guangzhou, China
B.S. in Mathematics, Minor in Economics 09/2019 - 06/2023
 - **Major GPA:** 3.9/4 (89/100)
 - **Undergraduate Thesis:** On the Minkowski sum of lattice points in lattice polytopes
 - **Selected Coursework:** Advanced Language Programming (C++); Data Structure and Algorithms; Mathematical Experiments and Mathematical Software; Database System; Operation Research.

RESEARCH INTERESTS

- **Pure mathematics:** toric geometry; tropical geometry; syzygies.
- **Applied mathematics:** human-computer interaction, neuroscience and mathematics.

PUBLICATIONS AND PREPRINTS

- Lei Song, Huanqi Wen, Zhixian Zhu. (2024). **On Covering Simplices by Dilations in Dimensions 3 and 4**, arXiv preprint, [arXiv:2404.02495](https://arxiv.org/abs/2404.02495)

RESEARCH EXPERIENCE

- **Positivity of Line Bundles on Toric Varieties** 09/2023 - present
Supervised by Prof. Lei Song
 - **Problem:** motivated by Mukai's conjecture, study whether the pair (X, L) of a projective toric variety X and an ample line bundle corresponding to a polytope P has the N_p properties.
 - **Topics:** stability of syzygy bundles, derived category, GIT theory, resolutions of toric subvarieties, Klyachko's classification of toric vector bundles.
 - **Computational experiments::** Used Macaulay2 and SageMath to experiment with syzygies, Hilbert functions, and examples of toric embeddings
- **Minkowski Sum of Lattice Points in Lattice Polytopes** 09/2022 - 09/2023
Supervised by Prof. Lei Song
 - **Problem:** motivated by Oda's conjecture, study which lattice polytopes are normal, specifically whether the lattice points in an $n + 1$ dilation of a polytope P can be written as the sum of a lattice point in P and a lattice point in an n dilation of P .
 - **Results:** proved that simplices with large lattice length in dimensions 3 and 4 are normal by constructing coverings with dilated simplices.
 - **Computer programming:** implemented Python and MATLAB programs to design and test algorithms for checking the covering property and normality of lattice polytopes.
- **Guangdong Digital Talent Training Program** 07/2022 - 08/2022
Internships at China Telecom.
 - gaining exposure to cloud computing and basic data analysis workflows

AWARDS AND SCHOLARSHIPS

- **Graduate Scholarship** 2023, 2024, 2025 *three times*
Sun Yat-sen University
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** 2022
Organized by the Consortium for Mathematics and Its Applications (COMAP)
- **Undergraduate Scholarship** 2020, 2021, 2022 *three times*
Sun Yat-sen University

ACADEMIC TALKS

- **Graded Syzygies** 2025 Spring
Syzygies Seminar, Sun Yat-sen University [\[notes\]](#)
- **A Series of Lectures on Toric Varieties and Relevant Classical Papers** 2024 Spring - 2024 Fall
Toric Geometry Seminar, Sun Yat-sen University [\[notes\]](#)
- **Introduction to Schemes** 2024 Spring - 2025 Spring
Algebraic Geometry Seminar, Sun Yat-sen University

CONFERENCES PARTICIPATION

- **2025 Summer Research Institute in Algebraic Geometry**, Colorado State University, USA 07/2025
- **Stable Reduction of Algebraic Curves (Mini-course)**, Xiamen University, China 05/2025
- **China Southeastern Algebraic Geometry Symposium VII**, Sun Yat-sen University, China 12/2024
- **2024 International Summer School of Algebraic Geometry**, Fudan University, China 07/2024

TEACHING EXPERIENCE

- **Graduate Teaching Assistant of Linear Algebra**, Sun Yat-sen University 2025 Fall

OTHER EXPERIENCE

- **International Scientific Research and Development Training** 2025 Spring
Completed the course on "Constructive Mathematics and Theoretical Computer Science," taught by Professor Vladimir Chernov.
- **Department Student Union Member and Head of a Division** 09/2019 - 06/2021
Planned and led diverse student activities within the department

SKILLS

- Languages: English (IELTS: 7.0), Mandarin (native)
- Programming: C++, Python
- Mathematical tools: MATLAB, SageMath, Macaulay2, $\LaTeX$